

Modelling Financial Markets

Chapter 20 (special chapter): Path Signatures and Realised Volatility — a Research Case Study



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Does a Method Built for Freight Rates Improve Volatility Forecasts?

Path Shape and Volatility Forecasts

- A research case study: the method of Gu et al. (2024), transferred to daily realised variance
  - the method was published for freight rates at Amazon
  - data: 50 assets from VOLARE (VOLatility Archive for Realized Estimates) Cipollini et al. (2026)
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

What you should be able to do

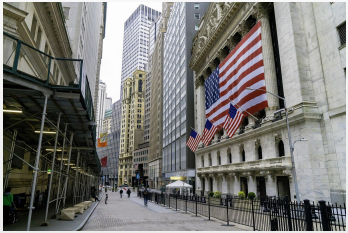
- compute the signature of a piecewise-linear path and read its Lévy area
- build signature features and kernel weights for a volatility forecast
- compare forecasts on many assets
  - with Diebold–Mariano (DM) tests and model confidence sets
  - with Holm and Benjamini–Hochberg (BH) corrections
- pre-register a method transfer and report a null result with its mechanism

Why Forecast Realised Variance?

- Volatility is an input to almost every risk measure in this course
  - Tail-risk measures (Chapter 7), option prices (Chapter 12), portfolio weights (Chapter 4)
- Intraday data make daily variance almost observable
  - the measure is realised variance (RV, Chapter 9)
  - So volatility forecasting becomes a regression problem with a noisy but model-free target
- The benchmark is hard to beat
  - the HAR (heterogeneous autoregressive) model Corsi (2009) (Chapter 9)
  - Three averages of past RV; linear; estimated by OLS (ordinary least squares) in a second
- Any new method must beat HAR out of sample, on many assets, with a proper test

Two Ingredients to Test: Signature Features and Kernel Weights

- Gu et al. describe a recent window of a series as a **path** and summarise it by its **signature**
  - Signature features enter a sparse linear regression (LASSO, least absolute shrinkage and selection operator, in two steps)
  - Past periods whose path looks like today's get more weight (a signature kernel)
- They report a more than fivefold accuracy gain over an industry forecast of freight rates
  - Question: does the same recipe improve on HAR-type models for realised variance?
- Each ingredient is tested separately, so that a gain can be attributed to the features or to the weights



Wall Street and the New York Stock Exchange, 25 March 2020

Photo: Billie Grace Ward (2020); CC BY 2.0; Wikimedia Commons

Realised Variance and Quadratic Variation (1/2)

- **Question**: what exactly do we forecast, and how do intraday data measure the variance of a day?
- The price model: the log price moves by a drift, a diffusion term and jumps

$$dp_t = \mu_t\, dt + \sigma_t\, dW_t + dJ_t$$

  - p_t: log price at time t (time in days); μ_t: drift; σ_t: spot volatility; W_t: Brownian motion; J_t: pure-jump process, ΔJ_s = jump size at time s
- Realised variance: the sum of squared intraday returns of day t Barndorff-Nielsen & Shephard (2002)

$$\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2 \xrightarrow{M\to\infty} \int_{t-1}^{t} \sigma_s^2\, ds + \sum_{t-1<s\le t} (\Delta J_s)^2$$

  - M: number of intraday returns in day t (for a US stock, 78 five-minute returns in the 6.5-hour session); $r_{t,i}$: the i-th intraday log return of day t
  - The limit is the quadratic variation of day t: the integrated variance plus the sum of squared jumps

Realised Variance and Quadratic Variation (2/2)

- Without jumps and noise, on an equally spaced grid, the limit is the integrated variance
  - the error is mixed Normal

$$\sqrt{M}\,(\mathrm{RV}_t - \mathrm{IV}_t) \Rightarrow \mathrm{MN}(0,\ 2\,\mathrm{IQ}_t)\,, \qquad \mathrm{IV}_t = \int_{t-1}^{t} \sigma_s^2\, ds, \quad \mathrm{IQ}_t = \int_{t-1}^{t} \sigma_s^4\, ds$$

  - IV_t: integrated variance, the true variance of day t; IQ_t: integrated quarticity
  - $\mathrm{MN}(0, V)$: mixed Normal, a Normal with a random variance V; given the volatility path, the error is Normal with variance $2\,\mathrm{IQ}_t/M$
- Reading: RV is a noisy measurement of IV_t
  - the noise is largest on volatile days (where IQ_t is large)
  - This is the measurement error that HARQ, the HAR model with realised quarticity, corrects for (next section)
- Microstructure noise makes very fine sampling biased
  - five-minute returns are the standard compromise (Chapter 9)

Semivariances and Quarticity in VOLARE

- Semivariances Barndorff-Nielsen, Kinnebrock & Shephard (2010): RV split by the sign of the intraday returns

$$\mathrm{RV}_t^+ = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} > 0\}, \qquad \mathrm{RV}_t^- = \sum_{i=1}^{M} r_{t,i}^2 \, 1\{r_{t,i} < 0\}$$

  - RV_t^+: variance from up-moves, RV_t^-: variance from down-moves
  - They add up: $\mathrm{RV}_t = \mathrm{RV}_t^+ + \mathrm{RV}_t^-$
- Realised quarticity: an estimate of IQ_t, the scale of the measurement error of RV

$$\mathrm{RQ}_t = \frac{M}{3} \sum_{i=1}^{M} r_{t,i}^4 \;\to\; \mathrm{IQ}_t$$

  - The factor $M/3$: a Normal return with variance s^2 has fourth moment $3s^4$, and each five-minute variance is about IV_t/M
  - The limit needs a continuous, noise-free price and an equally spaced grid; a jump ΔJ adds about $M(\Delta J)^4/3$, so raw RQ is jump-sensitive

▸ Appendix: five-minute RV, data

Targets and Losses (1/2): the Target

- The target at horizon h is the average RV over the next h days

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \frac{1}{h}\sum_{j=\mathbf{1}}^{h} \mathrm{RV}_{t+j}, \qquad h \in \{1, 5, 22\}$$

  - t: the forecast origin (the last day observed); $h = 1, 5, 22$: one day, one week, one month of trading days
- Log models forecast $y = \ln \mathrm{RV}_{t+\mathbf{1}:t+h}$ and return to levels with a variance correction

$$F_t = \exp(\hat{y}_t + \hat{s}^{\mathbf{2}}/2)$$

  - F_t: forecast of $\mathrm{RV}_{t+\mathbf{1}:t+h}$ made at t; $\hat{y}_t$: fitted log forecast; $\hat{s}^{\mathbf{2}}$: variance of the regression residuals
  - $+\hat{s}^{\mathbf{2}}/2$: since $\mathbb{E}[e^Z] = e^{\mu+s^{\mathbf{2}}/\mathbf{2}}$ for $Z \sim N(\mu, s^{\mathbf{2}})$; exact if the log error is conditionally Normal with constant variance, otherwise an approximation
- For $h > 1$ targets overlap: this induces serial dependence
  - through lag $h - 1$ by construction, possibly longer
  - so every test uses a HAC (heteroskedasticity and autocorrelation consistent) variance

Matching a Forecast to Its Target

- In the target definition, the subscript t is the forecast origin
- In loss and test formulas, the subscript labels a matched evaluation row
  - the realised target
  - the forecast made before that target became observable
- One-day horizon: information through Monday close forecasts Tuesday RV
  - stored next to the Tuesday realisation
- Five-day horizon: the same origin forecasts the average from Tuesday to the following Monday
  - the next origin gives an overlapping target
- Forecast comparison tests use the same realised target for both models in each row
- The row label is not permission to use that target in estimation

Targets and Losses (2/2): Robust Losses

- A loss is **robust** Patton (2011)
  - for a conditionally unbiased proxy, the *expected*-loss ranking with RV_t equals that with the true variance
  - Conditionally unbiased: $\mathbb{E}_{t-1}[\mathrm{RV}_t] = \sigma_t^2$, with $\mathbb{E}_{t-1}$ the expectation given information up to $t-1$ and σ_t^2 the true variance of day t
- The two robust losses used here (written for $h = 1$; for $h > 1$ replace RV_t by the target)

$$\mathrm{MSE}_t = (\mathrm{RV}_t - F_t)^2, \qquad \mathrm{QLIKE}_t = \frac{\mathrm{RV}_t}{F_t} - \ln \frac{\mathrm{RV}_t}{F_t} - 1 \;\geq\; 0$$

  - MSE (mean squared error): the squared error; QLIKE (quasi-likelihood loss) is zero only if $F_t = \mathrm{RV}_t$ and grows faster when the forecast is too low than when it is too high
  - Example with $\mathrm{RV}_t = 1$: $F_t = 0.5$ gives $\mathrm{QLIKE}_t = 2 - \ln 2 - 1 = 0.307$, $F_t = 2$ gives $0.5 - \ln 0.5 - 1 = 0.193$
- QLIKE depends on the ratio RV_t / F_t
  - the same proportional error has the same loss on a calm day as on a stormy day
- MSE is dominated by a few crisis days
  - QLIKE is the primary loss, MSE the secondary one

The Panel: 50 Assets in Three Classes

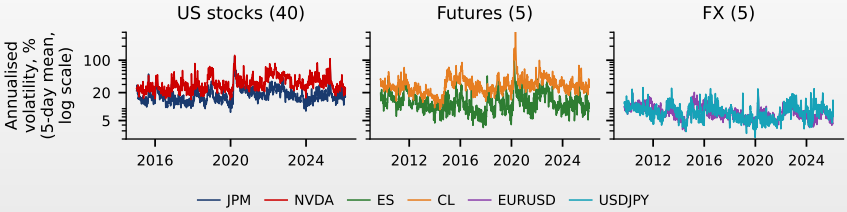
Class	Assets	Sample	Observations	Median volatility
American large-cap stocks	40	2 Jan 2015 – 30 Jan 2026	111 440	17.1%
Futures (C, CL, ES, GC, NG)	5	28 Sep 2009 – 30 Jan 2026	21 059	19.7%
FX (AUD, EUR, GBP, CAD, JPY vs USD)	5	25 Sep 2009 – 30 Jan 2026	21 203	7.3%

- **Question**: on which data do we test the transfer, and why on three asset classes?
- Assets
  - Stocks: 40 members of the Dow Jones 30 and S&P 100, labelled by ticker, from AAPL (Apple) to XOM (Exxon Mobil)
  - Futures: C = corn, CL = crude oil, ES = E-mini S&P 500, GC = gold, NG = natural gas
  - FX (foreign exchange): Australian dollar, euro, British pound, Canadian dollar and Japanese yen against the US dollar

The Panel: Median Volatility and Variables

- Three asset classes
  - a gain that appears in one class only does not show that the method transfers
- Median volatility in the table: $\sqrt{252\,\mathrm{median}(\mathrm{RV}_t)}$ per asset, then the median over the class
  - 252 trading days a year turn a daily variance into an annual one
  - The data contain trading days only
- Variables
  - five-minute RV, semivariances RV^+, RV^-, realised quarticity (RQ)
  - daily open and close prices

Realised Volatility Across Asset Classes (1/2)



- Annualised volatility $\sqrt{252\,\mathrm{RV}_t}$ in %, five-day mean, log scale, two assets per class
  - stocks: JPM = JPMorgan Chase, NVDA = Nvidia
  - futures: ES = E-mini S&P 500, CL = crude oil
  - FX: EURUSD = US dollars per euro, USDJPY = yen per US dollar
- All classes share March 2020; crude oil (CL) peaks in April 2020
  - a forecaster must adapt within days
- CL: the continuous VOLARE series joins successive contracts
  - the negative price of 20 April 2020 concerned only the May contract, close to expiry
  - the series stays positive

Realised Volatility Across Asset Classes (2/2)

- On a log scale volatility is persistent (the long memory HAR approximates)
- Spikes decay within weeks, so the shape of the recent path may carry information

MFM_ch20_case_study

HAR: Three Averages of the Past (1/2)

- **Question**: which simple benchmark must be beaten before we believe a new method?
- The HAR model Corsi (2009): future RV is a linear function of the daily, weekly and monthly averages of past RV

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \beta_d \,\mathrm{RV}_t + \beta_w \,\mathrm{RV}_t^{(w)} + \beta_m \,\mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

$$\mathrm{RV}_t^{(w)} = \frac{1}{5}\sum_{j=\mathbf{0}}^{\mathbf{4}} \mathrm{RV}_{t-j}, \qquad \mathrm{RV}_t^{(m)} = \frac{1}{22}\sum_{j=\mathbf{0}}^{\mathbf{21}} \mathrm{RV}_{t-j}$$

- Notation
  - $\mathrm{RV}_t^{(w)}$, $\mathrm{RV}_t^{(m)}$: weekly (5-day) and monthly (22-day) averages of RV up to day t; ε_{t+h}: forecast error
  - $\beta_{\mathbf{0}}$: intercept; $\beta_d, \beta_w, \beta_m \geq 0$ in practice: how much yesterday, last week and last month weigh; estimated by OLS for each horizon h
  - $\beta_d + \beta_w + \beta_m$ close to 1: high persistence of volatility
- Traders at daily, weekly and monthly frequencies
  - a cascade that mimics long memory with three parameters

HAR: Three Averages of the Past (2/2)

- HAR is a restricted AR(22): the 22 lag coefficients take only three distinct values

$$\phi_{\mathbf{1}} = \beta_d + \frac{\beta_w}{5} + \frac{\beta_m}{22}, \qquad \phi_{\mathbf{2..5}} = \frac{\beta_w}{5} + \frac{\beta_m}{22}, \qquad \phi_{\mathbf{6..22}} = \frac{\beta_m}{22}$$

  - ϕ_j: the coefficient of $\mathrm{RV}_{t+\mathbf{1}-j}$ in an AR(22) (autoregression with 22 lags)

- log-HAR: the same regression in logs

$$\ln \mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \beta_d \ln \mathrm{RV}_t + \beta_w \ln \mathrm{RV}_t^{(w)} + \beta_m \ln \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - No negative forecasts, less driven by spikes
  - Logs of averages are not averages of logs, so this log-HAR is not an exact restricted AR(22) in $\ln \mathrm{RV}$ (Seminar 20, A7)

▸ Appendix: HARQ and SHAR

Seven Models in One Table

Model	Target	Weights	Regressors
HAR	RV	equal	$\mathrm{RV}_t, \mathrm{RV}_t^{(w)}, \mathrm{RV}_t^{(m)}$
log-HAR	$\ln \mathrm{RV}$	equal	the same, in logs
HARQ	RV	equal	HAR + $\mathrm{RV}_t \left(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}}\right)$
SHAR	RV	equal	$\mathrm{RV}_t^+, \mathrm{RV}_t^-, \mathrm{RV}_t^{(w)}, \mathrm{RV}_t^{(m)}$
log-HAR-K	$\ln \mathrm{RV}$	signature kernel	log-HAR regressors
Sig-L	$\ln \mathrm{RV}$	equal	log-HAR + 36 non-constant signature terms (39 before dropping constants), two-step LASSO
Sig-LK	$\ln \mathrm{RV}$	signature kernel	as Sig-L: the method of Gu et al.

- Names: L = LASSO, K = kernel weights
  - the charts call Sig-L "Signature LASSO", Sig-LK "Signature LASSO + kernel weights", log-HAR-K "log-HAR + kernel weights"
- log-HAR, log-HAR-K, Sig-L and Sig-LK form a 2×2 ablation
  - each ingredient switched on and off separately
  - signature features (yes/no) $\times$ kernel weights (yes/no)
- Log-HAR is the natural baseline for the signature models
  - same target, and its regressors are among the candidates
  - but the LASSO may also drop log-HAR terms
  - so Sig-L vs log-HAR compares two procedures, not only two feature sets

A Window of Data Is a Path

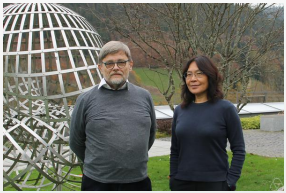
- **Question**: how do we summarise a 22-day window so that the order of the moves is kept, not only their levels?
- Take the last L days of $d-1$ series
  - add time as the first channel (time augmentation)
  - connect the points linearly

$$X : [0,1] \to \mathbb{R}^d, \qquad X_{u_i} = \left(u_i,\ \ln \mathrm{RV}_{t-L+1+i},\ \textstyle\sum_{j \le t-L+1+i} r_j\right), \qquad u_i = \tfrac{i}{L-1},\ i = 0, \dots, L-1$$

  - L: window length in days (here 22); d: number of channels (here $d = 3$: time, log RV, cumulative return); $u_i \in [0,1]$: rescaled time of the i-th point; $r_j = \ln(C_j/O_j)$: open-to-close log return of day j, with O_j, C_j the open and close prices
- A regression on the window sees 22 lags; a path summary sees their order, speed and co-movement
  - Example: RV rose while prices fell, or RV rose first and prices fell later

Where Signatures Come From

- Kuo-Tsai Chen, geometry of loop spaces
  - Chen (1957): the iterated integrals of a path form an algebra
  - Chen (1958): they determine the path up to reparametrisation and retraced pieces
- Lyons (1998): rough path theory
  - the signature is the natural feature set for controlled differential equations
  - A response to a path depends on it (to first order) only through its signature
- Machine learning: Levin, Lyons & Ni (2013), Chevyrev & Kormilitzin (2016), Lyons & McLeod (2022); finance: signature-based asset price models Cuchiero, Gazzani & Svaluto-Ferro (2023)



Terry Lyons (left), Oberwolfach, 2024

Photo: Katrin Schmid (2024); CC BY-SA 2.0 DE; Wikimedia Commons

The Signature: Definition (1/2)

- Why iterated integrals
  - level 1 keeps only the total increments
  - integrals over ordered times keep the order in which the channels move
- One signature term is an iterated integral of the path along a word of channels

$$S(X)_{a,b}^{i_{\mathbf{1}}\cdots i_k} = \int_{a<u_{\mathbf{1}}<\cdots<u_k<b} dX_{u_{\mathbf{1}}}^{i_{\mathbf{1}}} \cdots dX_{u_k}^{i_k}$$

- Notation
  - $X : [a, b] \to \mathbb{R}^d$: a path of bounded variation (finite total length), e.g. piecewise linear; X_u^i: its channel i at time u
  - Word $(i_{\mathbf{1}}, \ldots, i_k)$: an ordered list of k channels, letters in $\{1, \ldots, d\}$; k is the **level** of the term
  - $u_{\mathbf{1}} < \cdots < u_k$: the integral runs over ordered times; dX_u^i: increment of channel i at time u (for a linear segment, slope times du)

The Signature: Definition (2/2)

- The signature is the whole sequence of levels

$$\mathrm{Sig}(X) = (1,\ S^{(\mathbf{1})},\ S^{(\mathbf{2})},\ S^{(\mathbf{3})}, \dots), \qquad S^{(k)} \in (\mathbb{R}^d)^{\otimes k}$$

  - $S^{(k)}$: the d^k terms of level k, one for each word of length k, stored as a k-dimensional array (a tensor; $\otimes$: tensor product)
  - $\mathrm{Sig}^N(X)$: the truncated signature, levels 0 to N; N is the depth
- Level 1: the total increments of the channels

$$S^i = X_b^i - X_a^i$$

- Level 2: increments of channel j weighted by how far channel i has already moved

$$S^{ij} = \int_a^b (X_u^i - X_a^i)\, dX_u^j$$

  - Each level adds one integral and records finer details of the order of the moves

Linear Segments and Chen's Identity (1/2)

- A linear segment with increment $\Delta \in \mathbb{R}^d$ has the tensor exponential as signature

$$\mathrm{Sig}(\text{segment}) = \exp_{\otimes}(\Delta) = \left(1,\ \Delta,\ \tfrac{1}{2!}\Delta^{\otimes \mathbf{2}},\ \tfrac{1}{3!}\Delta^{\otimes \mathbf{3}}, \dots\right)$$

  - $\Delta = X_b - X_a$: the increment of the segment; $\Delta^{\otimes k}$: the array of all products $\Delta^{i_\mathbf{1}} \cdots \Delta^{i_k}$, so a level-$k$ term is $\Delta^{i_\mathbf{1}} \cdots \Delta^{i_k}/k!$

- **Chen's identity** Chen (1957)
  - the signature of two paths run one after the other is the tensor product of their signatures

$$\mathrm{Sig}(X * Y) = \mathrm{Sig}(X) \otimes \mathrm{Sig}(Y), \qquad (A \otimes B)^{(k)} = \sum_{p=\mathbf{0}}^{k} A^{(k-p)} \otimes B^{(p)}$$

  - $X * Y$: concatenation, Y starts where X ends; $A^{(k)}$, $B^{(k)}$: level k of the signatures A, B, with $A^{(\mathbf{0})} = B^{(\mathbf{0})} = 1$
  - Level k of the whole path collects every way of splitting a word of length k between the first and the second piece

Linear Segments and Chen's Identity (2/2)

- A piecewise-linear path with m segments, increments $\Delta_1, \ldots, \Delta_m$

$$\mathrm{Sig} = \exp_{\otimes}(\Delta_1) \otimes \cdots \otimes \exp_{\otimes}(\Delta_m)$$

  - This is the whole algorithm: m updates, each costing $O(d^N)$ at depth N
- Rolling windows: Chen also removes an initial piece

$$\mathrm{Sig}(X_{[s,t]}) = \mathrm{Sig}(X_{[a,s]})^{-1} \otimes \mathrm{Sig}(X_{[a,t]})$$

  - $X_{[s,t]}$: the path restricted to $[s,t]$, with $a \leq s \leq t$; $(\cdot)^{-1}$: inverse for $\otimes$, the signature of the piece run backwards

Worked Example: a Two-Segment Path

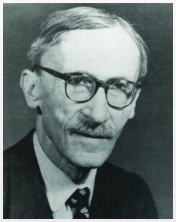
- The path $X : (0,0) \to (1,0) \to (1,2)$, with increments $\Delta_1 = (1,0)$ and $\Delta_2 = (0,2)$
- Level 1: $S^{(1)} = \Delta_1 + \Delta_2 = (1,2)$
- Level 2 by Chen's identity: $S^{ij} = \frac{1}{2}\Delta_1^i \Delta_1^j + \Delta_1^i \Delta_2^j + \frac{1}{2}\Delta_2^i \Delta_2^j$
  - Δ_m^i is the component i of the increment of segment m
- Result: $S^{11} = \frac{1}{2}$, $S^{12} = 0 + 1 \cdot 2 + 0 = 2$, $S^{21} = 0$, $S^{22} = \frac{1}{2} \cdot 2^2 = 2$
- Check: $S^{11} = (\Delta X^1)^2/2$ and $S^{22} = (\Delta X^2)^2/2$
  - shuffle identity: the product of two level-1 coordinates equals the sum of their two orderings at level 2
- $S^{12} \neq S^{21}$: the order matters, since channel 1 moves in the first segment and channel 2 in the second
- An average of the points cannot tell this path from the one that goes up first and right second
  - the signature can

The Lévy Area: Order as a Number

- The Lévy area A^{ij}: the antisymmetric part of level 2

$$A^{ij} = \tfrac{1}{2}(S^{ij} - S^{ji})$$

  - Equals the signed area between the path, projected on channels (i, j), and its chord (the straight line from start to end)
  - $A^{ij} > 0$: channel i tends to move before channel j
- Symmetric part: $\tfrac{1}{2}(S^{ij} + S^{ji}) = \tfrac{1}{2}\,\Delta X^i\,\Delta X^j$, no new information
  - Worked example: $A^{\mathbf{12}} = (2 - 0)/2 = 1$, the area of the triangle between the path and the chord
- Channels (return, log RV): a price fall followed by a volatility rise gives $A < 0$
  - so does a price rise followed by a volatility fall
- A lead–lag statistic
  - leverage-type sequences (Chapter 5) are one contributor to a negative area
  - the sign alone does not identify them



Paul Lévy (1886–1971)

Photo: Konrad Jacobs; CC BY-SA 2.0 DE; Wikimedia Commons

The Signature Kernel

- Kernel = inner product of the truncated signatures Király & Oberhauser (2019): a similarity score between two paths

$$k_N(X, Y) = \langle \mathrm{Sig}^N(X), \mathrm{Sig}^N(Y) \rangle = 1 + \sum_{k=1}^{N} \sum_{|w|=k} S(X)^w S(Y)^w$$

  - X, Y: two paths with the same channels; w: a word, $|w|$: its length (level); $S(X)^w$: the term of word w; the leading 1 is level 0
- Distance induced by the kernel (Gu et al., Eq. 9)
  - squared Euclidean distance between the truncated signatures

$$d_{\mathrm{Sig}}(X, Y) = k_N(X, X) - 2\, k_N(X, Y) + k_N(Y, Y) = \|\mathrm{Sig}^N(X) - \mathrm{Sig}^N(Y)\|^2$$

  - $d_{\mathrm{Sig}} = 0$: the two windows have the same signature up to depth N; large d_{Sig}: different shapes
- Gu et al. and this chapter use this **truncated** kernel, with N as for the features
  - The untruncated kernel solves a Goursat PDE (partial differential equation) Salvi et al. (2021); the paper cites it but does not use it
- Channels are rescaled before the kernel (Appendix): otherwise the channel with the largest units dominates

The Problem at Amazon

- **Question**: what exactly did the paper do, and how strong is its evidence?
- Weekly North American truckload rates, national and five regions, 2018–2022 Gu et al. (2024)
  - forecast up to 12 weeks ahead
  - More than 100 candidate factors screened to about 40
  - In Amazon operations since November 2022; KDD 2024 (Knowledge Discovery and Data Mining, the conference of the ACM, Association for Computing Machinery)
- One direct regression per horizon Δt (in weeks), $y_{\tau+\Delta t}$ on the features ξ_τ (next slide)



A container truck on an American interstate, 2025

Photo: DanTD (2025); CC BY 4.0; Wikimedia Commons

The Features of the Paper

- Features at week τ (Section 4.1, Eq. 14)
  - external factors x_τ
  - the signature of the time-augmented target path

$$\xi_\tau = \left[x_\tau,\ \mathrm{Sig}^N(\hat{y}_{\tau-l:\tau})\right], \qquad \hat{y}_{\tau-l:\tau} = \{(t_i, y_{t_i})\}$$

  - y_{t_i}: the rate in week t_i, $\tau - l \leq t_i \leq \tau$
  - l: lookback in weeks
  - $\hat{y}_{\tau-l:\tau}$: the time-augmented path of the last l weeks
- The external factors bring the level and information from outside the series
  - the signature brings the shape of the recent path

The Method: Two-Step LASSO

- Step 1, selection by a weighted LASSO Tibshirani (1996) (Eq. 14)

$$\hat{\theta}_{\mathrm{L}} = \arg\min_{\theta} \sum_{\tau} w_{\tau} \left(y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta\right)^{2} + \lambda \|\theta\|_{1}$$

  - θ: coefficient vector, $\|\theta\|_{1} = \sum_{j} |\theta_{j}|$; $w_{\tau} \geq 0$: weight of week τ (equal, or from the kernel, next slide)
  - $\lambda \geq 0$: penalty; a larger λ sets more coefficients exactly to zero; $\lambda = 0$ is weighted OLS
- Step 2, refit by weighted OLS on the selected support only, the nonzero coefficients of step 1 (Eq. 15)

$$\hat{\theta} = \arg \min_{\mathrm{supp}(\theta) = \mathrm{supp}(\hat{\theta}_{\mathrm{L}})} \sum_{\tau} w_{\tau} \left(y_{\tau+\Delta t} - \xi_{\tau}^{\top} \theta\right)^{2}$$

- Why refit: LASSO shrinks the kept coefficients towards zero
  - OLS after selection removes that bias Belloni & Chernozhukov (2013)
  - The paper calls it "adaptive" because of the weights w_{τ}, not in the sense of the adaptive LASSO of Zou (2006)
- Choice of λ in this chapter: the support with the smallest BIC Efron et al. (2004)
  - BIC: Bayesian Information Criterion
  - along the LARS (Least Angle Regression) path
  - in each training window (Appendix)

The Method: Kernel Weights

- Distance of each past window to the last window with an observed target (Eq. 11)

$$\delta_\tau = d_{\mathrm{Sig},N}\left(z_{\tau-l:\tau},\ z_{t-\Delta t-l:t-\Delta t}\right)$$

  - $z_\tau = (x_\tau, y_{\tau+\Delta t})$: the *pairs* of predictors and later target, not the features ξ_τ (Algorithm 1); $z_{\tau-l:\tau}$: their path over l weeks
  - t: the current week; $t-\Delta t$: the last week whose target y_t is already observed; $d_{\mathrm{Sig},N}$: the signature distance at depth N
- Softmax weights with temperature $\gamma \geq 0$ (Eq. 12): similar windows (small δ_τ) get more weight

$$w_\tau = \frac{\exp(-\gamma\,\delta_\tau)}{\sum_s \exp(-\gamma\,\delta_s)}$$

  - s: runs over all training weeks, so the weights sum to 1; $\gamma = 0$: ordinary (equal-weight) regression; $\gamma \to \infty$: only the most similar past window
- A nearest-neighbour idea inside a regression: learn from the past periods that looked like today
  - Cost of the idea: fewer effective observations, $n_{\mathrm{eff}} = 1/\sum_\tau w_\tau^2$ (Kish); $n_{\mathrm{eff}} = n$ for equal weights $1/n$, $n_{\mathrm{eff}} = 1$ if one week takes all the weight

What the Paper Reports

- National rates, three months ahead, April–November 2021 (Table 1)
  - relative error $|y - \hat{y}|/|y|$ (y: realised rate, $\hat{y}$: its forecast)
  - about 1–2% vs 13–27% for the industry forecast, "more than fivefold"
  - Horizons 1–12 weeks (Table 2): 1.06–2.60% nationally, 1.05–5.69% by region; estimated annual saving of about 50 million dollars
  - Kernel ablation (Table 3, six weekly values, 31 October–5 December 2021, one region): errors up to 9.85% without the kernel vs up to 5.15% with it

What the Paper Reports: Limits of the Evidence

- **One benchmark**: an unnamed commercial forecast
  - no ARIMA (autoregressive integrated moving average) model
  - no LASSO without signatures, no random walk
- **Short evaluation**: a few months, a few series, no standard errors, no test of equal accuracy
- **Undisclosed tuning**: l, N, γ, λ and how they were chosen
- None of this makes the method wrong; it makes the evidence thin
  - which is why it is tested elsewhere
  - replication before extension (Chapter 19)

What Transfers to Volatility: Kept and Adapted

Transfers

- Signature of the time-augmented recent path of the target
- Two-step LASSO with weights, direct regression per horizon
- Softmax weights $\exp(-\gamma\,\delta_\tau)$ with the truncated signature kernel

Must be adapted

- Daily, noisy target instead of weekly rates: logs, HAC inference, robust losses
- External factors x_τ replaced by the log-HAR terms (they carry the level)
- The kernel compares the paths (time, $\ln \mathrm{RV}$, return) ending at τ and at t
  - updated every 5 days
  - not the pairs $(x_\tau, y_{\tau+\Delta t})$ ending at $t - \Delta t$
- Undisclosed choices (l, N, γ, λ) fixed ex ante
  - strong benchmarks and formal tests replace the single commercial forecast

Four Hypotheses Fixed Before the Results

- **Question**: what do we fix before the results, so that a gain cannot be found by searching?
- **H1 (features)**: signature terms add information beyond log-HAR
  - Test: Sig-L vs log-HAR, same target, same equal weights
- **H2 (weights)**: kernel weights improve a given regression
  - Tests: Sig-LK vs Sig-L and log-HAR-K vs log-HAR
- **H3 (method)**: the full method of Gu et al. beats the HAR family
  - Sig-LK vs HAR and vs log-HAR
  - model confidence set, MCS, of all seven models
- **H4 (stress)**: any gain is larger in stress periods
  - COVID-19 crash, 2–30 April 2025, the US tariff shock
  - where similar past windows should matter most

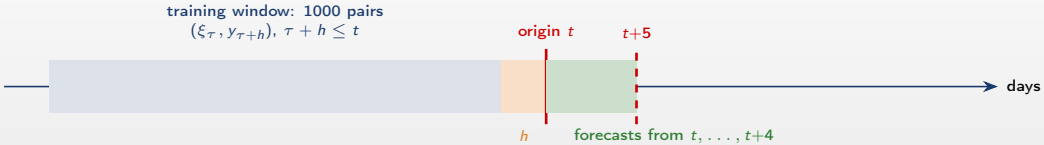
Kernel Temperature: One Choice per Class, on a Validation Block

- Scale-free temperature, set in every training window

$$\gamma = \frac{c}{\mathrm{median}_\tau(\delta_\tau)}$$

  - δ_τ: kernel distances of the training days to the current window; $c > 0$: a dimensionless constant, the only tuning parameter
  - $c = 1$: a window at the median distance gets weight e^{-1} relative to an identical one; larger c: more concentrated weights
- Grid $c \in \{0.5, 1, 2, 4\}$, chosen per horizon and asset class
  - on the first 250 forecast days of each asset, pooled within the class
  - Criterion: mean over the class of QLIKE(c) divided by the asset's average over the grid
  - Evaluation starts after the last validation target of the class, so no tuning target postdates an evaluated origin
- For the 40 stocks the smallest temperature, $c = 0.5$, wins at every horizon for Sig-LK
  - the most nearly equal weights
  - for futures and FX the choice changes with the horizon
  - a flat, noisy criterion (all values in the Appendix)

Rolling Estimation, Step by Step



- Re-estimate every 5 trading days on the last 1000 complete pairs
  - between re-estimations the coefficients are fixed
  - the regressors are updated daily
- Kernel distances are recomputed at each re-estimation, against the window ending at the origin t

Evaluation Protocol (1/3): the DM Test

- **Question**: is the loss difference between two models larger than sampling noise?
- Per asset and horizon: mean QLIKE and MSE Diebold & Mariano (1995)
  - compared by the DM (Diebold–Mariano) test of equal predictive accuracy
  - the paper's relative error is reported descriptively only (Appendix)

$$d_t = L(\mathrm{RV}_t, F_t^A) - L(\mathrm{RV}_t, F_t^B), \qquad \mathrm{DM} = \frac{\bar{d}}{\sqrt{\hat{\Omega}/T}}$$

  - L: the loss (QLIKE or MSE); F_t^A, F_t^B: forecasts of models A and B; d_t: their loss differential
  - $\bar{d}$: the mean of d_t over the T evaluation days; $\hat{\Omega}$: the long-run variance of d_t (Further Topics)
- Under $H_0 : \mathbb{E}[d_t] = 0$, DM is approximately $N(0, 1)$
  - $\mathrm{DM} < -1.645$ means model A is significantly more accurate at 5% (one-sided)
- Rolling estimation, so the test is read as the unconditional GW (Giacomini–White) test
  - The conditional GW test also asks whether d_{t-h} predicts d_t (instruments $(1, d_{t-h})$) Giacomini & White (2006)

Appendix: HAC variance

Evaluation Protocol (2/3): the Model Confidence Set

- Why a set of models: the DM test compares two models
  - here there are seven and no benchmark chosen in advance
- MCS at 90% Hansen, Lunde & Nason (2011): the set of models that contains the best one with probability at least 90%
  - Built by elimination: while the $T_{\max}$ statistic (the largest standardised gap between a model's mean loss and the average over the models still in the set) rejects equal accuracy, drop the worst model
  - p-values by the stationary bootstrap Politis & Romano (1994), 1000 replications, mean block length $\max(h, 10)$ days
  - The loss series are resampled jointly, to keep the dependence across models

Evaluation Protocol (3/3): Panel Test and Corrections

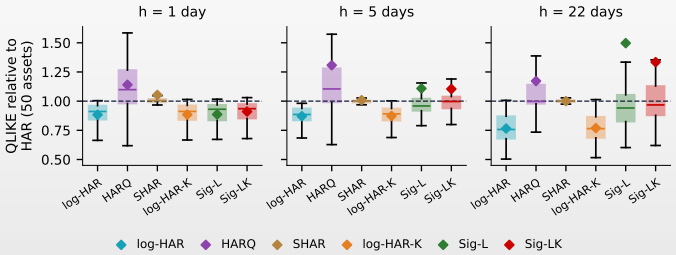
- Panel: DM test on the cross-sectional mean loss differential (one series, one value per date)
- Count the assets (of 50) with significantly better results
  - The Holm correction Holm (1979) controls the FWER (family-wise error rate, the probability of at least one false rejection)
  - BH Benjamini & Hochberg (1995) controls the FDR (false discovery rate, the expected share of false rejections among the rejections)
- Subperiods (evaluation periods by class: Appendix)
  - 20 Feb–30 Apr 2020 (COVID-19 crash)
  - 2–30 Apr 2025 (US tariff shock)
  - all other days

Appendix: Holm, BH, MCS examples

A Disclosed Deviation

- Pre-registered: the BPQ insanity filter on all models
  - A test run of the code on JPM (one asset, before any panel result) showed it replacing correct crash forecasts by the calm in-sample mean
  - On 12 March 2020 the realised next-day RV of JPM was about 40 times the filtered HAR forecast
- Changed before estimating the models on the 50-asset panel
  - primary results with positivity only
  - a forecast ≤ 0 becomes the in-sample mean
  - The BPQ filter is kept as a robustness check and reported in full

QLIKE Relative to HAR, Asset by Asset



- **Question**: does the method of Gu et al. beat the HAR models and, above all, log-HAR?
- Ratio for each asset: mean QLIKE of the model over mean QLIKE of HAR; below 1: more accurate than HAR
  - Boxes: interquartile range over the 50 assets; line: median; diamond: mean; whiskers: 1.5 interquartile ranges
- Ratios are averaged over assets with equal weights
  - here, in the ablation and in the filtered results
  - the mean of the ratios is not the ratio of the means
  - so a new benchmark means new per-asset ratios
- Medians describe a typical asset; means remain sensitive to a few large losses

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Reading the QLIKE Comparison

- $h = 1$: log-HAR, log-HAR-K and Sig-L are close (0.883, 0.885, 0.886)
  - Features alone (Sig-L vs log-HAR): +0.4% in QLIKE, so 36 extra features bring no gain
  - Features and weights (Sig-LK vs log-HAR): +4.5%
- $h = 5, 22$: Sig-LK falls behind log-HAR (+28.0%, +92.9%)
  - Its medians relative to HAR (0.997, 0.968) are much better than its means: a few assets and days drive the loss
- HARQ and SHAR lose to HAR in levels here
  - because the main design does not apply the BPQ filter
  - the filter rescues HARQ (Appendix)
- First verdict: the method does not beat the simplest log model
  - a near-tie is not a proof of equal accuracy

Asset by Asset: Wins and Losses, Corrected

Comparison (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	2/0/0 | 5/0/0	0/0/0 | 15/0/2	0/0/0 | 27/1/16
Sig-L vs log-HAR	3/1/1 | 5/0/0	0/0/0 | 9/0/0	1/0/0 | 21/0/0
log-HAR-K vs log-HAR	3/0/0 | 3/0/0	3/0/0 | 1/0/0	5/0/0 | 6/0/0
Sig-LK vs Sig-L	2/0/0 | 5/0/0	0/0/0 | 5/0/0	2/0/0 | 10/0/0
Sig-LK vs HAR	12/5/6 | 0/0/0	7/1/1 | 0/0/0	8/2/4 | 1/0/0
log-HAR vs HAR	21/5/17 | 0/0/0	23/13/20 | 0/0/0	42/25/42 | 0/0/0

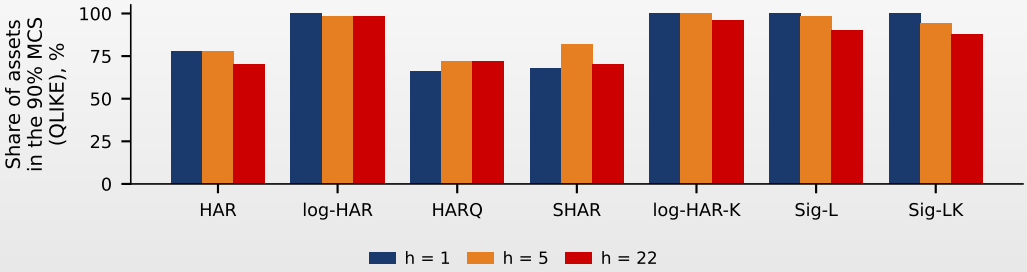
- Cell: significantly better on raw/Holm/BH assets | significantly worse on raw/Holm/BH assets
  - one-sided DM at 5%, family of 50 assets
  - Holm controls the FWER under any dependence
  - BH controls the FDR under independence or PRDS (positive regression dependence on a subset) Benjamini & Yekutieli (2001)

Asset by Asset: Reading the Table

- H1 (signatures): Sig-L beats log-HAR on 3 of 50 assets at $h = 1$ raw, 1 after Holm
  - about 2.5 raw wins are expected by chance
- Also at $h = 1$, Sig-L loses to log-HAR on 5 assets raw and on none after Holm
- H2 (weights): log-HAR-K vs log-HAR and Sig-LK vs Sig-L: no Holm-significant difference at any horizon
- Against HAR at $h = 1$, Sig-LK wins on 12 assets, but log-HAR alone wins on 21
  - choose the benchmark that isolates the idea

Appendix: DM statistics

Model Confidence Sets (1/2)



- Share of the 50 assets in which each model belongs to the 90% MCS under QLIKE ($h = 1, 5, 22$)
  - log-HAR 100%, 98%, 98%
  - Sig-LK 100%, 94%, 88%
- The MCS rarely separates the four log models
  - HAR, HARQ and SHAR are excluded for a fifth to a third of the assets
  - the level target is the clearest loser

Model Confidence Sets (2/2)

- Under MSE almost every model stays in the set
  - mean-loss differences have large standard errors

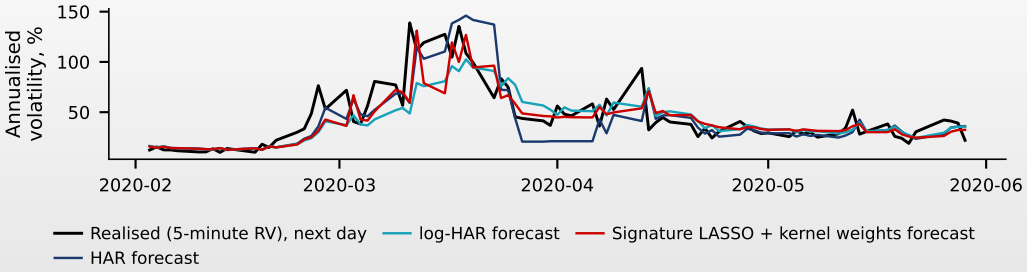
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Panel Tests: Pooling the Evidence

Comparison: DM t (one-sided p)	$h = 1$	$h = 5$	$h = 22$
Sig-LK vs log-HAR	+1.45 (0.927)	+1.68 (0.953)	+1.80 (0.964)
Sig-L vs log-HAR	-1.91 (0.028)	+1.48 (0.931)	+1.51 (0.935)
log-HAR-K vs log-HAR	+1.89 (0.971)	+0.40 (0.654)	+1.92 (0.973)
Sig-LK vs Sig-L	+1.74 (0.959)	+0.98 (0.837)	-0.77 (0.221)
log-HAR vs HAR	-3.09 (<0.001)	-3.95 (<0.001)	-4.50 (<0.001)

- Series: the mean QLIKE differential over all assets trading that day
  - negative t = first model better
- Pooled, only the log transform is clearly significant
- Sig-L vs log-HAR at $h = 1$: significant in a single one-sided test ($p = 0.028$), driven by futures
  - For the 40 stocks alone the sign flips: $t = +1.72$ against Sig-L
- After a Holm correction over the three horizons (threshold $0.05/3 \approx 0.017$) it is no longer significant
- Conditional GW test, Sig-LK vs log-HAR: 0, 0 and 0 Holm rejections out of 50

Inside the COVID-19 Crash: JPM, Next Day (1/2)



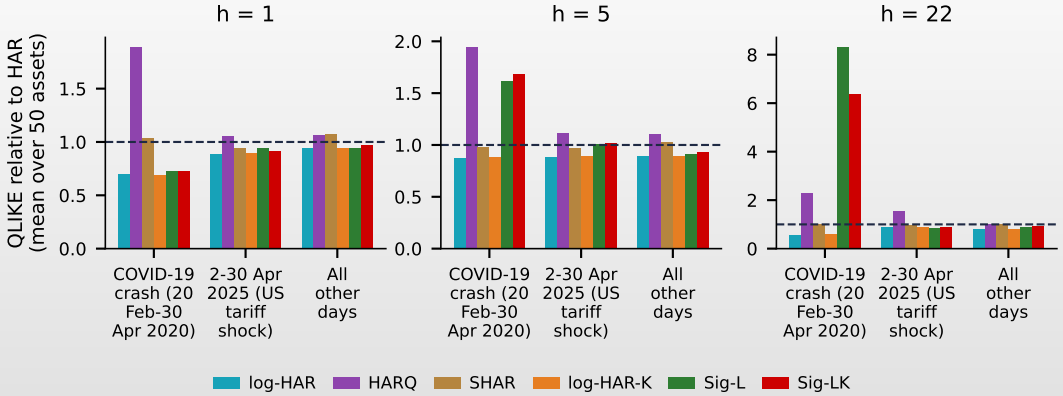
- Annualised volatility in % of the forecasts, $\sqrt{252 F_t}$, and of the target, $\sqrt{252\,\mathrm{RV}_{t+\mathbf{1}}}$
  - February–May 2020
  - all models lag the first jump
- Sig-LK reacts slightly faster than log-HAR in mid-March
  - when the path has already turned
  - after the peak both decay at similar speeds

Inside the COVID-19 Crash: JPM, Next Day (2/2)

- HAR in levels overshoots, then turns negative in late March
  - the positivity rule replaces it by the in-sample mean (the flat segment)

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Stress Periods vs Normal Times (H4) (1/2)



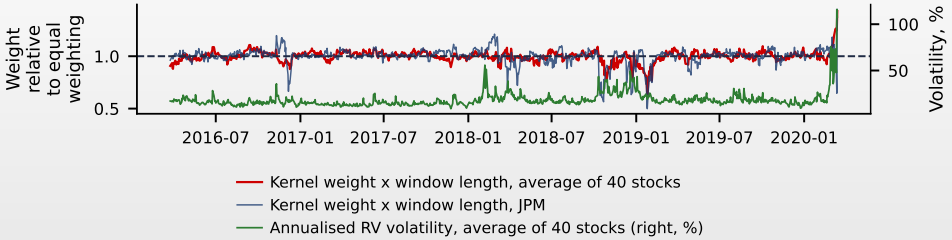
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Stress Periods vs Normal Times (H4) (2/2)

- Mean over assets of QLIKE relative to HAR within each subperiod fixed before the evaluation
- $h = 1$, COVID-19 crash: log-HAR 0.70, Sig-LK 0.73, HARQ 1.89
  - 2–30 April 2025 (US tariff shock): 0.88 and 0.92
  - other days: 0.94 and 0.97
- $h = 22$, COVID-19 crash: Sig-LK 6.35 and Sig-L 8.30 times the loss of HAR
  - exactly where similar past windows should help, the signature models fail most
  - H4 not supported (descriptive, no formal interaction test)

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Kernel Weights at the Origin 16 March 2020 (1/2)



- Weights of Sig-LK ($h = 1$) at the origin 16 March 2020 over the training days
  - training days: 24 Mar 2016 – 13 Mar 2020
  - weights times the window length: 1 = equal weighting
- Nearly equal: between 0.61 and 1.43 times $1/n$
  - effective sample size 975 of 1000 (average over the 40 stocks)

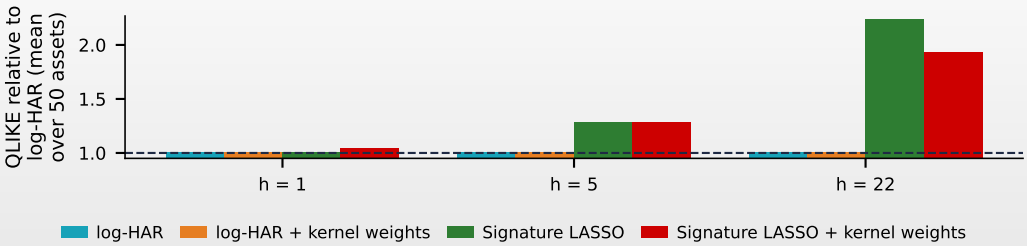
Kernel Weights at the Origin 16 March 2020 (2/2)

- The highest weights go to the last days before the origin
- Recent 22-point windows overlap the current window
  - similarity may partly reflect overlap, not a distinct economic regime
- The late-2018 sell-off gets *less* weight
  - weights are almost uncorrelated with log volatility (+0.04)
  - the translation-invariant signature sees shape, not level

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Appendix: stress weights, LASSO

The 2×2 Ablation



- **Question**: which ingredient drives the result, the signature features or the kernel weights?
- Mean over the 50 assets of QLIKE relative to log-HAR
  - features (signature or not) crossed with weights (kernel or equal)
- The dashed line marks 1 (log-HAR)
  - a bar above it means a worse forecast than log-HAR

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The 2×2 Ablation: the Numbers

- Weights alone (log-HAR-K): 1.002, 1.002, 1.004 at $h = 1, 5, 22$: no effect
- Features alone (Sig-L): 1.004, 1.285, 2.238
- Both (Sig-LK): 1.045, 1.280, 1.929
  - Average effective sample size with the Sig-LK weights: 816 of 1000
  - Adding the weights to Sig-L reduces the loss at $h = 22$ without removing it
  - They worsen $h = 1$ through crude oil (CL: 2.71), unlike the kernel gain in Table 3 of Gu et al.

Appendix: robustness of Sig-LK

By Asset Class

Sig-L relative to log-HAR (QLIKE)	$h = 1$	$h = 5$	$h = 22$
Stocks (40)	1.009	1.332	2.505
Futures (5)	0.965	1.084	1.206
FX (5)	0.996	1.111	1.134

- The only gain: futures at $h = 1$ (-3.5%) and, marginally, FX
  - Five futures and one horizon: a candidate for a new, pre-registered study, not a finding of this one
- Stocks, 80% of the panel, show no gain: the pooled result is mostly a stock result

Limitations of This Study

- One design out of many: a null result for this design is not one for signatures in general
  - the design: $L = 22$, $N = 3$, three channels, BIC, a four-point grid for c
- Truncated kernel only
- No external factors (implied volatility, macro news), central in Gu et al.
- Realized GARCH Hansen, Huang & Shek (2012) and machine-learning benchmarks are outside the comparison
  - GARCH: generalised autoregressive conditional heteroskedasticity (Chapter 5)
  - machine-learning benchmarks: Chapter 13
- Stocks are evaluated over six years only

Open Directions

- A level-aware kernel: basepoint augmentation or the level of $\ln \mathrm{RV}$ in the distance (Seminar 20, C2)
- Signatures of the intraday path (5-minute returns)
- The PDE kernel Salvi et al. (2021)
- Log-signatures and bounded (tanh) channels against explosive level-3 terms Morrill et al. (2020)
- Cross-asset paths, where Lévy areas measure lead–lag across markets
- Futures at $h = 1$ on a longer panel, with this design fixed in advance

When Does a Method Transfer Across Domains?

- Methods travel faster than evidence
  - a KDD paper on freight rates becomes a candidate for volatility, credit, energy
  - Which properties of the source problem made the method work?
  - Does the target share them?
- Finance adds a hard filter Menkveld et al. (2024)
  - 164 teams on the same data disagree more than their standard errors suggest
- Criterion: a transfer succeeds if the gain over the strongest simple benchmark survives
  - a pre-registered set $\mathcal{S}$ of designs, assets and horizons
  - with the family corrected
  - In this chapter: $\mathcal{S} = 50$ assets $\times$ 3 horizons $\times$ 4 variants; Holm/BH were applied per comparison and horizon (families of 50 assets), variants reported descriptively; no joint correction over all 600 cells

The Discovery Loop Applied to This Chapter

- **Read**: an assistant lists every equation, hyperparameter and benchmark of the paper
  - it marks what is not disclosed
  - each item is checked against the PDF (prompts)
- **Map**: for each component, does it transfer, what must change, which benchmark isolates it
- **Pre-register**: hypotheses, grid, tests and correction
  - fixed in a dated document before the first result
  - the assistant drafts, the team decides every number
- **Build and test the pieces**: signatures against identities, not against another library
  - level 1 = increment, Chen on a split path
- **Run the grid, then attack it**
  - Holm over 50 assets
  - a search for the single asset that would make a misleading headline (prompts)
- **Explain the null**, checked with data (checklist: Appendix)
  - here flat weights (translation invariance)
  - explosive level-3 terms (exponential link)

Mini-Case: Three Headlines From the Same Runs

- "Signatures beat HAR on 12 of 50 assets" (one-day horizon, raw DM)
  - True, but log-HAR alone beats HAR on 21: the gain belongs to the log transform
- "Signatures improve E-mini S&P 500 volatility forecasts" (ES, $h = 1$, DM $t = -2.91$ against log-HAR)
  - One cell out of 150 asset-horizon cells: after Holm, 0 of 50 at $h = 1$ and none at $h = 5, 22$
- The headline of the pre-registered evaluation
  - "no gain over log-HAR; worse at longer horizons; kernel weights inert"
  - An AI assistant asked to "find where the method works" will produce the first two; the plan protects the third

Your Turn: Design a Transfer Study

- Pick one open direction of this chapter
  - level-aware kernel, intraday signatures, log-signatures, futures only
  - Write H_0/H_1, the benchmark that isolates it, and the result that would make you drop it
- Use an assistant to list what could go wrong in your design, then write one unit test per item
- Fix the pre-registration in a dated document before the first out-of-sample number
  - run the grid
  - report the distribution over $\mathcal{S}$
- Log every AI error you catch in `AI_ERRORS.md` (Seminar 20, C1 gives typical ones)

Key Takeaways

- Signatures turn a window into a feature vector with a theory behind it
  - Chen's identity, uniqueness, universality
- Transferred to realised variance, the method of Gu et al. does not beat log-HAR
  - H1 (signatures add information): not supported, no gain at one day, a loss at a week and a month
  - H2 (kernel weights help): not supported, weights are almost flat and down-weight past stress; H3: only HAR in levels is beaten, through the log target; H4 (gains in stress): not supported
- The translation-invariant kernel sees shape, not level
  - level-3 terms explode in crashes
- The strongest effect in the study is the oldest one: model the log of RV
- The null result is informative
  - the ablations and the mechanism explain why the method does not transfer

Readings on Signatures and Realised Measures

Reading

- Signatures: Chevyrev & Kormilitzin (2016); Lyons & McLeod (2022)
  - realised measures: Franke, Härdle & Hafner (2019) and Chapter 9

Advanced courses on quantinar.com

- Statistics of Financial Markets
- Applied Time Series Analysis with Python
- Machine Learning in Financial Risk

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_20
  - data: VOLARE, free for research at volare.unime.it

Check your understanding

- What information about order can a path signature retain that a simple average loses? (the two-segment example)
- Why compare with both a level target and a log target?
- Which ablation isolates the contribution of signatures? (the 2×2 ablation)
- Can one significant win among many assets establish superiority?
- What is the role of the correction for multiple comparisons? (evaluation protocol)
- How can a null result still be informative? (limitations and open directions)

Why Five-Minute RV, and Where Do the Data Come From?

- Alternatives to RV
  - bipower variation (robust to jumps)
  - the realised kernel (robust to noise) Barndorff-Nielsen et al. (2008)
  - range-based estimators
- Target of this chapter: five-minute RV
  - across 400 assets, little beats it as a proxy of daily variance Liu, Patton & Sheppard (2015)
- Its semivariances and quarticity are on the same grid
  - so SHAR (the semivariance HAR model) and HARQ use consistent inputs
- Every model is evaluated on the same target
  - so loss differences come from the models, not from the volatility measure
- VOLARE is built at the Universities of Messina and Florence from tick data Cipollini et al. (2026)
  - with the same estimators for every asset class
- Free for research at volare.unime.it, not redistributed with this course
  - download the realised-variance files
  - point the notebooks to your local folder
- The authors request the citation
  - Cipollini, Cruciani, Gallo, Insana, Otranto & Spagnolo (2026), arXiv:2602.19732

Two Refinements (1/2): HARQ

- RV measures the true variance with error

$$\mathrm{RV}_t = \mathrm{IV}_t + \eta_t, \qquad \mathrm{Var}(\eta_t) \approx 2\,\mathrm{IQ}_t / M$$

  - η_t: measurement error; errors-in-variables attenuate (pull towards zero) β_d, most on noisy days
- HARQ Bollerslev, Patton & Quaedvlieg (2016): the weight on yesterday's RV falls when RV is measured less precisely

$$\mathrm{RV}_{t+\mathbf{1}:t+h} = \beta_{\mathbf{0}} + \left(\beta_d + \beta_Q(\mathrm{RQ}_t^{\mathbf{1/2}} - \overline{\mathrm{RQ}^{\mathbf{1/2}}})\right)\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - $\mathrm{RQ}_t^{\mathbf{1/2}}$: square root of realised quarticity, a measure of the error of RV_t; $\overline{\mathrm{RQ}^{\mathbf{1/2}}}$: its in-sample mean, so β_d is the weight on an average day
  - Expected $\beta_Q < 0$: on a noisy day (RQ_t above average) yesterday's RV gets less weight
- BPQ (Bollerslev–Patton–Quaedvlieg) filter ("insanity filter")
  - a forecast outside the in-sample range of the target is replaced by the in-sample mean

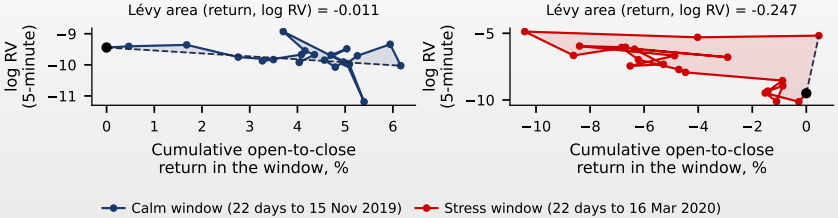
Two Refinements (2/2): SHAR

- SHAR Patton & Sheppard (2015): split the daily term by the sign of the intraday returns

$$\mathrm{RV}_{t+1:t+h} = \beta_0 + \beta^+ \mathrm{RV}_t^+ + \beta^- \mathrm{RV}_t^- + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+h}$$

  - RV_t^+, RV_t^-: the positive and negative semivariances of day t
  - $\beta^+ = \beta^-$ gives back HAR, since $\mathrm{RV}_t^+ + \mathrm{RV}_t^- = \mathrm{RV}_t$
- Patton and Sheppard find $\beta^- > \beta^+$
  - Variance from down-moves predicts more future variance than variance from up-moves: an intraday version of the leverage effect (Chapter 5)
- The signature can capture a related effect from daily data
  - the Lévy area between returns and log RV (Path Signatures section)

Two Windows of JPM as Paths (1/2)



- Path (cumulative open-to-close return, log RV) over 22 days
  - dashed chord from start to end
  - the shaded region gives the Lévy area
- Calm window (to 15 Nov 2019) vs stress window (to 16 Mar 2020)
  - calm: log RV moves by -0.58, the path wanders, Lévy area -0.011
  - stress: log RV rises by +4.32 (RV $\times$ 75) while the intraday return path falls and then recovers
  - stress: Lévy area -0.247 (returns in decimal units for the area, percent on the axis)

Two Windows of JPM as Paths (2/2)

- The net intraday return is only $+0.5\%$
  - the crash was largely overnight, outside this channel
  - no signature level recovers what the input omits

MFM_ch20_signatures

Why Signatures Work (1/2): Uniqueness and Shuffles

- **Uniqueness** Hambly & Lyons (2010): the full signature identifies the path
  - $\mathrm{Sig}(X)$ determines X up to tree-like equivalence (adding pieces that are retraced backwards) and translation
  - With a strictly increasing time channel nothing can be retraced, so only translation remains
  - A finite truncation Sig^N is not injective: two different paths can share levels $1..N$
- **Shuffle identity**: a product of signature terms is a sum of higher terms

$$S^i S^j = S^{ij} + S^{ji}, \qquad S^{ij} S^k = S^{ijk} + S^{ikj} + S^{kij}$$

  - The sum runs over all interleavings (shuffles) of the two words that keep the order inside each word
  - So a linear model in $\mathrm{Sig}^N(X)$ already contains polynomials of the increments

Why Signatures Work (2/2): Universality and Decay

- **Universality**: any continuous function of the path is approximately linear in the signature
  - Stone–Weierstrass, via the shuffle identity

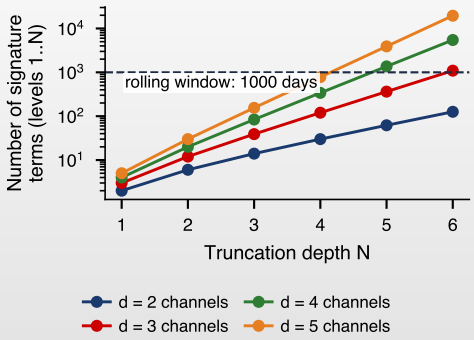
$$\forall \varepsilon > 0 \;\; \exists\, \ell \text{ linear}: \quad \sup_{X \in K} \left| f(X) - \langle \ell, \mathrm{Sig}(X) \rangle \right| < \varepsilon$$

  - K: a compact set of time-augmented paths with a common start (in the 1-variation topology); $f : K \to \mathbb{R}$: a continuous function of the path
  - $\langle \ell, \mathrm{Sig}(X) \rangle$: a weighted sum of finitely many signature terms, i.e. a linear regression on them
  - Sufficient depth permits nonlinear approximation; this does not guarantee accuracy at depth 3 or out of sample. The cost is d^N features, so sparsity is needed
- **Factorial decay**: high levels are small

$$\|S^{(k)}\| \leq \frac{\|X\|_{\text{1-var}}^k}{k!}$$

  - $\|X\|_{\text{1-var}}$: total length of the path (sum of the lengths of its segments); $\|S^{(k)}\|$: size (Euclidean norm) of level k
  - Truncating at a small depth N loses little for paths of moderate length, which is why channels are rescaled (Appendix)

How Many Features? The Truncation Dimension



- Number of terms in levels $1..N$ of a d-dimensional path (a geometric sum, since level k has d^k words)
  - $\dim(d, N) = \sum_{k=1}^{N} d^k = \frac{d^{N+1}-d}{d-1}$
- This chapter: $d = 3$ channels (time, log RV, cumulative return), depth $N = 3$: $3 + 9 + 27 = 39$ terms
  - Three are constant (words with only the time letter), so 36 vary
- $d = 5$, $N = 4$ would give 780 terms for 1000 observations: the LASSO would select from noise
- Here $N = 3$ and $L = 22$ are fixed ex ante, with $N = 2$ as a robustness check

MFM_ch20_signatures

Two Invariances and the Basepoint

- **Translation invariance**: $\mathrm{Sig}(X + c) = \mathrm{Sig}(X)$ for any constant vector $c \in \mathbb{R}^d$
  - only increments enter
  - The signature does not see the **level** of volatility, so the level enters separately: the log-HAR terms play the role of the external factors x_τ of Gu et al.
- **Basepoint augmentation** Morrill et al. (2020): the path starts at the origin, then runs through the points of the window

$$\hat{X} = (0,\ X_{u_{\mathbf{0}}},\ X_{u_{\mathbf{1}}}, \ldots, X_{u_{L-\mathbf{1}}}), \qquad \mathrm{Sig}(\hat{X}) = \exp_{\otimes}(X_{u_{\mathbf{0}}}) \otimes \mathrm{Sig}(X)$$

  - The first segment has increment $X_{u_{\mathbf{0}}}$, so level 1 contains the starting point (for example $\ln \mathrm{RV}$ at the start of the window): the signature becomes level-aware
  - Translating the path to start at 0 does not do this: it is itself a translation (Seminar 20, C2 uses the prepended segment)
- **Reparametrisation invariance**: $X_{\psi(u)}$, with ψ increasing, has the same signature
  - so without a time channel a 22-day move and a 2-day move of the same shape look alike
  - the strictly increasing time channel $u_i = i/(L-1)$ removes this invariance
- In this chapter: time augmentation, no basepoint
  - so the kernel compares shapes, not levels
  - the consequence shows in the kernel weights (Results section)

Three Windows: Weights and Information

- Distances $\delta = (0.1, 0.4, 0.9)$ and $\gamma = 2$
- Unnormalised weights: $(0.819, 0.449, 0.165)$, with sum 1.433
- Normalised weights: $w = (0.571, 0.313, 0.115)$
- $n_{\text{eff}} = 1/\sum w_i^2 \approx 2.28$, compared with 3 under equal weights
- γ acts as an inverse temperature
  - a larger γ concentrates the weights, gaining similarity and losing information

Evaluation Protocol: the HAC Variance

- Why not the plain sample variance: d_t is autocorrelated
  - overlapping targets up to lag $h-1$
  - volatility clustering in the losses
  - so $\hat{\gamma}_0/T$ understates the uncertainty of $\bar{d}$ and over-rejects
- The Newey–West estimator Newey & West (1987), heteroskedasticity and autocorrelation consistent (HAC)

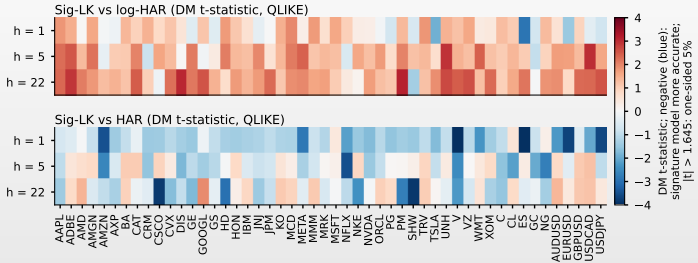
$$\hat{\Omega} = \hat{\gamma}_0 + 2\sum_{j=1}^{q}\Big(1-\frac{j}{q+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{T}\sum_{t=j+1}^{T}(d_t-\bar{d})(d_{t-j}-\bar{d})$$

  - $\hat{\gamma}_j$: the sample autocovariance of d_t at lag j
  - The Bartlett weights $1-j/(q+1)$ decline linearly with the lag and guarantee $\hat{\Omega}\geq 0$
  - $q=\max(h,\lceil T^{1/3}\rceil)$ lags: at least h, for the dependence created by the overlap ($\lceil\cdot\rceil$: rounding up)

Two Numerical Examples: Holm, BH and the MCS

- Holm compares three sorted p-values, $(0.004, 0.023, 0.08)$, in turn with $(0.05/3, 0.05/2, 0.05)$
  - the first two pass, the third fails, so we stop
- BH compares them with $(0.05/3, 2 \cdot 0.05/3, 0.05)$ and keeps the largest passing rank: again two
- In a family of 50 tests the same p-values may no longer pass
  - which is why the family is fixed before the results
- MCS with three models A, B and C, standardised loss gaps 2.4, -0.5 and -0.7
  - the bootstrap p-value is 0.04, so at 10% we remove A
- Recompute the test for B and C only: the new p-value is 0.30, so we stop and B and C remain
- Failure to separate them is not proof that their population losses are equal
- The 90% coverage is a repeated-sample guarantee, not a posterior probability for the set obtained

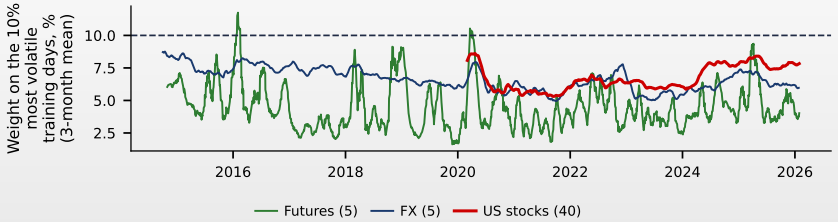
DM Statistics for Every Asset and Horizon



- Diebold–Mariano t-statistics of the QLIKE differential, one column per asset ticker
  - 40 stocks, 5 futures, 5 FX pairs
  - blue: Sig-LK more accurate, red: less accurate
- Against log-HAR (top): pale at $h = 1$, red at $h = 5, 22$
  - blue exceptions: ES, TSLA = Tesla, EURUSD
  - five assets significantly red before correction
- Against HAR (bottom): blue at $h = 1$ because Sig-LK inherits the log target

MFM_ch20_case_study

Does the Kernel Ever Favour Stress? (1/2)



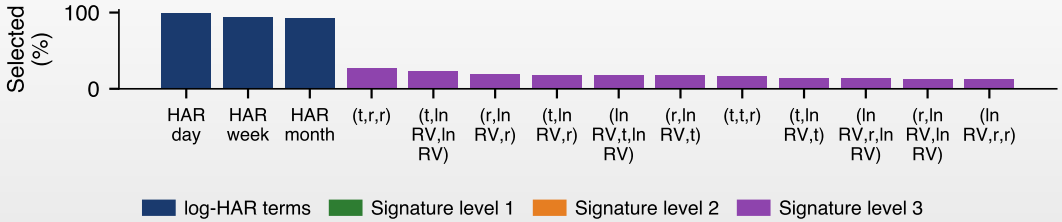
- Weight that Sig-LK ($h = 1$) puts on the 10% most volatile days of its training window
  - averaged by class, three-month mean
- Stocks, per re-estimation: mean 6.7%, below the 10% of equal weighting
  - between 4.4% and 9.9% (maximum on 17 Mar 2020)
- FX stays below 10% throughout
- Futures exceed 10% (three-month mean) only in two short episodes
  - early 2016 (up to 11.7%)
  - March–April 2020

Does the Kernel Ever Favour Stress? (2/2)

- The mechanism behind the null result for H2
  - the weights do not implement "learn from past crises"

MFM_ch20_case_study

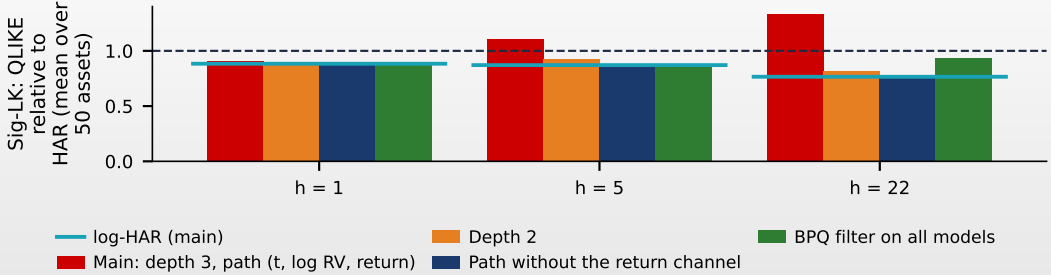
What the LASSO Selects



- Across all assets and horizons: share of re-estimations in which a regressor is in the support of Sig-LK
  - the support is selected by BIC
  - labels: HAR day/week/month = the three log-HAR terms, t = time, ln RV = $\ln \mathrm{RV}$, r = cumulative return
- The log-HAR terms are kept in 99%, 94% and 93% of the re-estimations
  - the LASSO rediscovers HAR
  - average support 8.7 regressors (Sig-L), 6.7 (Sig-LK)
- The most frequent signature term, (t,r,r) (27%), and all frequent ones are of level 3
  - cubic in the increments
  - far outside the training range in a crash, amplified by $\exp(\cdot)$

MFM_ch20_case_study

Robustness of Sig-LK



- Sig-LK relative to HAR (mean over 50 assets)
  - in the main design and in two pre-registered variants
  - with the BPQ filter (the disclosed deviation)
- The reference line marks log-HAR
  - a bar below 1 means a more accurate forecast than HAR

MFM_ch20_case_study

Robustness of Sig-LK: the Variants

- Depth 2: 0.922 and 0.820 at $h = 5, 22$ (main design: 1.104, 1.334)
- Path without returns: 0.878 and 0.770
  - so the long-horizon failures come from terms that contain the return channel
- Depth 2 also reduces them, so they come mostly from level-3 terms
- Without returns the model approaches log-HAR (0.871 and 0.765)
  - depth 2 trails it
- Both variants reduce instability
  - neither establishes H1

Features and Scaling

◂ Route ◂ Decay ◂ Kernel

- Regressors at day τ for the signature models: 3 log-HAR terms + 39 signature terms
  - $\xi_\tau = \left[\ln \mathrm{RV}_\tau,\ \ln \mathrm{RV}_\tau^{(w)},\ \ln \mathrm{RV}_\tau^{(m)},\ \mathrm{Sig}^3(X_{\tau-21:\tau})\right]$, path X = (time, $\ln \mathrm{RV}$, cumulative r), 22 points
- Each non-time channel q is multiplied by $\kappa_q = 1/(s_q\sqrt{L-1})$
  - so its typical total move over the window is about 1
  - s_q: standard deviation of the daily increments of channel q in the training window up to the origin t; $L-1 = 21$ increments per window
  - Time is not rescaled (constant increments, $s = 0$): it runs from 0 to 1; a word $(i_1..i_k)$ is rescaled by $\prod_j \kappa_{i_j}$
- Before the LASSO every column is standardised with weighted moments of the training window
  - constant columns (words of time only) are dropped
- Raw features at τ use data up to τ
  - scales and standardisation use data up to the origin t
  - the no-look-ahead cutoff is t

Estimation: Choosing λ Without Look-Ahead

- Inside each training window: the whole weighted LASSO path by LARS
  - For every support on the path: weighted OLS refit (step 2 of Gu et al.)
- Pick the support that minimises a BIC-type criterion with the Kish effective sample size
  - a heuristic of this design
  - $\mathrm{BIC}(S) = n_{\mathrm{eff}} \ln \hat{s}^2_S + (|S|+1) \ln n_{\mathrm{eff}}, \quad n_{\mathrm{eff}} = 1/\sum_\tau w_\tau^2$
  - S: a support (set of kept regressors), $|S|$: its size, $+1$ for the intercept; $\hat{s}^2_S$: weighted residual variance of the refit on S
- Why not cross-validation: 50 assets $\times$ 3 horizons $\times$ hundreds of re-estimations
  - BIC uses only the window
  - Kish size corrects for unequal weights, not for serial dependence (overlapping targets, $h > 1$): not a likelihood-based BIC for these data

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Evaluation Windows

Class	Evaluation starts	Ends	Days per asset (median)
Stocks	27 Jan 2020	29 Jan 2026	1 499
Futures	25 Sep 2014	29 Jan 2026	2 919
FX	27 Aug 2014	29 Jan 2026	2 952

- 265 929 forecasts per model over the three horizons: 89 893 ($h = 1$), 89 293 ($h = 5$), 86 743 ($h = 22$)
- Stocks start in 2015, so their evaluation begins in January 2020
  - the COVID-19 crash is in every stock sample
  - Futures and FX start in 2009 and are evaluated from 2014

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Chosen Kernel Temperatures

	Sig-LK			log-HAR-K		
Class	$h = 1$	$h = 5$	$h = 22$	$h = 1$	$h = 5$	$h = 22$
Stocks	0.5	0.5	0.5	0.5	0.5	1.0
Futures	2.0	1.0	0.5	2.0	1.0	4.0
FX	0.5	2.0	0.5	4.0	2.0	4.0

- The value of c from the grid $\{0.5, 1, 2, 4\}$, chosen on the validation block of each class and horizon

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The Paper's Metric on VOLARE

Median of asset mean errors $\lvert \mathrm{RV} - F \rvert / \mathrm{RV}$ (%)	$h = 1$	$h = 5$	$h = 22$
HAR	57.2	48.9	51.7
log-HAR	50.4	41.9	38.7
HARQ	55.0	49.0	50.7
SHAR	57.9	49.2	51.8
log-HAR-K	49.7	41.0	38.7
Sig-L	50.1	42.6	42.6
Sig-LK	50.6	43.6	43.7

- Median over assets of the mean relative error: about 40–60%
  - against 1–2% for freight rates in Gu et al.
- Daily RV is a noisy, heavy-tailed target
  - the evaluated models have much larger relative errors than those reported for freight rates
  - this does not establish a lower bound on attainable error
  - Sig-LK forecasts above ten times the in-sample maximum: 8 at $h = 1$, 29 at $h = 5$ (concentrated on 10 assets) and 36 at $h = 22$; log-HAR: none

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The Insanity Filter, Revisited

QLIKE relative to HAR, BPQ filter on all models	$h = 1$	$h = 5$	$h = 22$
HAR	1.000	1.000	1.000
log-HAR	0.763	0.654	0.778
HARQ	0.995	0.912	1.014
SHAR	1.014	1.003	1.000
log-HAR-K	0.772	0.659	0.779
Sig-L	0.871	0.819	0.904
Sig-LK	0.890	0.867	0.930

- The filter rescues HARQ relative to HAR (0.995 at $h = 1$ instead of 1.140), as BPQ intended
- Sig-LK relative to log-HAR, both filtered: 1.238, 1.424, 1.209
  - the gap widens at $h = 1, 5$ and narrows at $h = 22$
- Filtered forecasts, $h = 1$: HAR 180, HARQ 140, Sig-LK 143, log-HAR 29 (of 89 893)
  - the filter hits the signature models far more often than log-HAR

Back

What the Researcher Must Check

- **The method is the paper's**: kernel, weights, refit, as in the cited equations
  - Test: each component of the method points to the equation, table or section of the paper it comes from; undisclosed choices are labelled as choices of the study
- **No look-ahead**: λ, scales, c and features use only data available at the origin
  - Test: change every value after the origin (for example, multiply future RV by 10); the forecast made at the origin must not change
- **The benchmark isolates the idea**: the 2×2 ablation is reported, not only the full model against HAR
- **Losses and tests are the right way round**: $\mathrm{QLIKE}(\mathrm{RV}, \mathrm{RV}) = 0$ and $\mathrm{QLIKE}(\mathrm{RV}, F) > 0$ otherwise
  - The DM variance is HAC with bandwidth at least h

Back

The Discovery Loop, Prompts (1/2)

- **Read**: extract exactly what the paper did
  - "From Gu et al. (2024, KDD), list every equation of the estimator, every hyperparameter and its reported value, and the benchmark of each table. Mark what is not disclosed."
  - Check each item against the PDF: equation numbers, table numbers, the kernel actually used
- **Map**: what transfers, what must be adapted
  - "Our target is daily realised variance. For each component of the method, say whether it transfers, what must change, and which benchmark isolates it."
- **Pre-register**: hypotheses, grid, tests, correction
  - fixed in a dated document signed by the team before the first result

Back

The Discovery Loop, Prompts (2/2)

- **Build and test the pieces**: signatures against identities, not against another library
  - "Write unit tests for a depth-3 signature: level 1 equals the increment, the symmetric part of level 2 equals half the outer product, Chen's identity on a split path."
- **Run the grid, then attack it**
  - "Here are all 600 runs (50 assets $\times$ 3 horizons $\times$ 4 specifications: main, depth 2, no return channel, BPQ filter). Which conclusions survive Holm over 50 assets? Which single asset or subperiod would make a misleading headline?"
- **Explain the null**: find the mechanism, not an excuse

Back

Acronyms used in this material (1/2)

- **AAPL**: Apple Inc. (Nasdaq ticker)
- **ACM**: Association for Computing Machinery
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive model
- **ARIMA**: AutoRegressive Integrated Moving Average (model)
- **AUD**: Australian Dollar (ISO 4217 code)
- **BH**: Benjamini–Hochberg (procedure)
- **BIC**: Bayesian Information Criterion
- **BPQ**: Bollerslev–Patton–Quaedvlieg (insanity filter of the HARQ paper)
- **CAD**: Canadian Dollar (ISO 4217 code)
- **CL**: Crude oil (WTI light sweet) futures, NYMEX ticker
- **COVID-19**: COronaVIrus Disease 2019
- **DM**: Diebold–Mariano (test)
- **ES**: E-mini S&P 500 futures (CME ticker)
- **EURUSD**: Euro / US dollar exchange rate (FX pair code)
- **FDR**: False Discovery Rate
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GC**: Gold futures, COMEX ticker
- **GW**: Giacomini–White (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)

Acronyms used in this material (2/2)

- **HAR**: Heterogeneous AutoRegressive (model)
- **HARQ**: HAR model with Realised Quarticity
- **JPM**: JPMorgan Chase & Co. (NYSE ticker)
- **KDD**: Knowledge Discovery and Data Mining (ACM SIGKDD conference)
- **LARS**: Least-Angle RegreSsion (algorithm)
- **LASSO**: Least Absolute Shrinkage and Selection Operator
- **MCS**: Model Confidence Set
- **MSE**: Mean Squared Error
- **NG**: Natural Gas futures, NYMEX ticker
- **NVDA**: NVIDIA Corporation (Nasdaq ticker)
- **OLS**: Ordinary Least Squares
- **PDE**: Partial Differential Equation
- **PRDS**: Positive Regression Dependence on a Subset (of p-values)
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RQ**: Realised Quarticity
- **RV**: Realised Variance
- **SHAR**: Semivariance HAR model
- **TSLA**: Tesla, Inc. (Nasdaq ticker)
- **US**: United States
- **USDJPY**: US dollar / Japanese yen exchange rate (FX pair code)
- **VOLARE**: VOLatility Archive for Realized Estimates (database)
- **WTI**: West Texas Intermediate (crude oil benchmark)
- **XOM**: Exxon Mobil Corporation (NYSE ticker)

References (1/3)

- Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536.
- Barndorff-Nielsen, O.E., Kinnebrock, S., Shephard, N. (2010). Measuring downside risk: realised semivariance. In *Volatility and Time Series Econometrics*, Oxford University Press, 117–136.
- Barndorff-Nielsen, O.E., Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society B*, 64(2), 253–280.
- Belloni, A., Chernozhukov, V. (2013). Least squares after model selection in high-dimensional sparse models. *Bernoulli*, 19(2), 521–547.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 1165–1188.
- Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18.
- Chen, K.-T. (1957). Integration of paths, geometric invariants and a generalized Baker–Hausdorff formula. *Annals of Mathematics*, 65(1), 163–178.
- Chen, K.-T. (1958). Integration of paths—a faithful representation of paths by noncommutative formal power series. *Transactions of the American Mathematical Society*, 89(2), 395–407.
- Chevyrev, I., Kormilitzin, A. (2016). A primer on the signature method in machine learning. arXiv:1603.03788.
- Cipollini, F., Cruciani, G., Gallo, G.M., Insana, A., Otranto, E., Spagnolo, F. (2026). VOLatility Archive for Realized Estimates (VOLARE). arXiv:2602.19732.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.

References (2/3)

- Cuchiero, C., Gazzani, G., Svaluto-Ferro, S. (2023). Signature-based models: theory and calibration. *SIAM Journal on Financial Mathematics*, 14(3), 910–957.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- Efron, B., Hastie, T., Johnstone, I., Tibshirani, R. (2004). Least angle regression. *The Annals of Statistics*, 32(2), 407–499.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- Gu, H., Guo, X., Jacobs, T.L., Kaminsky, P., Li, X. (2024). Transportation marketplace rate forecast using signature transform. *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 4997–5005; arXiv:2401.04857.
- Hambly, B., Lyons, T. (2010). Uniqueness for the signature of a path of bounded variation and the reduced path group. *Annals of Mathematics*, 171(1), 109–167.
- Hansen, P.R., Huang, Z., Shek, H.H. (2012). Realized GARCH: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- Király, F.J., Oberhauser, H. (2019). Kernels for sequentially ordered data. *Journal of Machine Learning Research*, 20(31), 1–45.
- Levin, D., Lyons, T., Ni, H. (2013). Learning from the past, predicting the statistics for the future, learning an evolving system. arXiv:1309.0260.

References (3/3)

- Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- Lyons, T. (1998). Differential equations driven by rough signals. *Revista Matemática Iberoamericana*, 14(2), 215–310.
- Lyons, T., McLeod, A.D. (2022). Signature methods in machine learning. arXiv:2206.14674.
- Menkveld, A.J., Dreber, A., Holzmeister, F., Huber, J., Johannesson, M., Kirchler, M., et al. (2024). Nonstandard errors. *Journal of Finance*, 79(3), 2339–2390.
- Morrill, J., Fermanian, A., Kidger, P., Lyons, T. (2020). A generalised signature method for multivariate time series feature extraction. arXiv:2006.00873.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- Patton, A.J., Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3), 683–697.
- Politis, D.N., Romano, J.P. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Salvi, C., Cass, T., Foster, J., Lyons, T., Yang, W. (2021). The signature kernel is the solution of a Goursat PDE. *SIAM Journal on Mathematics of Data Science*, 3(3), 873–899.
- Tibshirani, R. (1996). Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society B*, 58(1), 267–288.
- Zou, H. (2006). The adaptive lasso and its oracle properties. *Journal of the American Statistical Association*, 101(476), 1418–1429.